

▼ **Figure 26.10 Monophyletic, paraphyletic, and polyphyletic groups.**

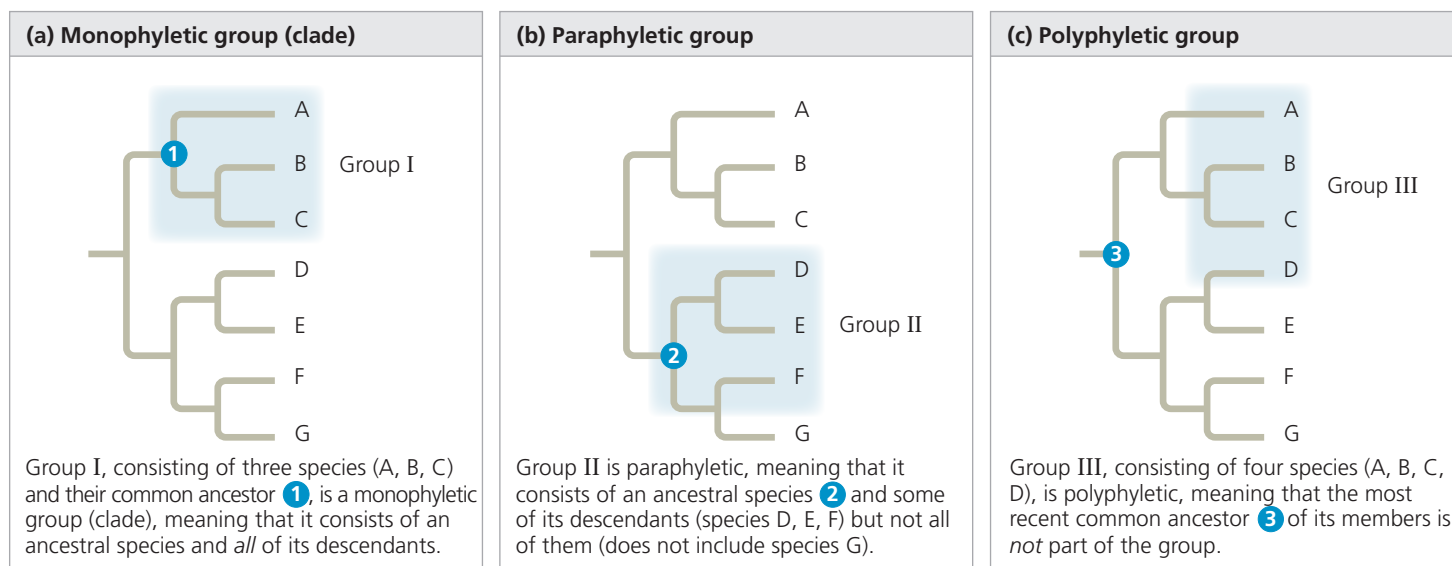


Figure 26.4, for example, the cat group (Felidae) represents a clade within a larger clade (Carnivora) that also includes the dog group (Canidae).

A taxon is equivalent to a clade only if it is **monophyletic** (from the Greek, meaning “single tribe”), signifying that it consists of an ancestral species and all of its descendants (**Figure 26.10a**). Contrast this with a **paraphyletic** (“beside the tribe”) group, which consists of an ancestral species and some, but not all, of its descendants (**Figure 26.10b**), or a **polyphyletic** (“many tribes”) group, which includes distantly related species but does not include their most recent common ancestor (**Figure 26.10c**).

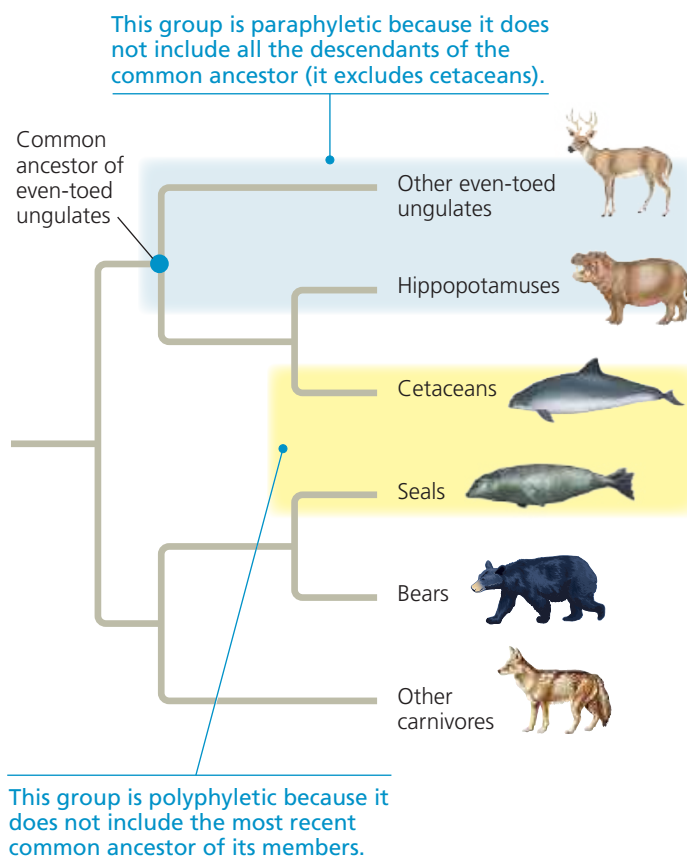
Note that in a paraphyletic group, the most recent common ancestor of all members of the group *is* part of the group, whereas in a polyphyletic group the most recent common ancestor *is not* part of the group. For example, a group consisting of even-toed ungulates (hippopotamuses, deer, and their relatives) and their common ancestor is paraphyletic because it does not include cetaceans (whales, dolphins, and porpoises), which descended from that ancestor (**Figure 26.11**). In contrast, a group consisting of seals and cetaceans (based on their similar body forms) is polyphyletic because it does not include the common ancestor of seals and cetaceans. Biologists avoid defining such polyphyletic groups; if new evidence indicates that an existing group is polyphyletic, its members are reclassified.

Shared Ancestral and Shared Derived Characters

As a result of descent with modification, organisms have characters they share with their ancestors, and they also have characters that differ from those of their ancestors. For example, all mammals have backbones, but a backbone does not distinguish mammals from other vertebrates because *all* vertebrates have backbones. The backbone predates the branching of mammals from other vertebrates. Thus for mammals, the backbone is a **shared ancestral character**, a character that originated in

an ancestor of the taxon. In contrast, hair is a character shared by all mammals but *not* found in their ancestors. Thus, in mammals, hair is considered a **shared derived character**, an evolutionary novelty unique to a clade.

▼ **Figure 26.11 Paraphyletic versus polyphyletic groups.**



DRAW IT Circle the branch point that represents the most recent common ancestor of cetaceans and seals. Explain why that ancestor would not be part of a cetacean–seal group defined by similar body form.